

Cyber & Information Security

Lab 2: overthewire Bandit



Linux Commands





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SSH

SSH (Secure Shell) is like a safe secret tunnel that lets you connect to another computer over the internet. Imagine you want to sit in front of a server in another country you don't need to travel, you just use SSH! It keeps your messages private and makes sure nobody can spy on them or change them.



How SSH Works?

1. Client and Server: Think of it as two sides of a phone call. You (the client) use your computer to call the server (the remote machine).
2. Encryption: Everything you type is scrambled (encrypted) before it travels, so even if someone is listening on the network, they'll only see nonsense.
3. Authentication: SSH checks who you are and who the server is. This can be done with a password or, more safely, with keys (like a private key you keep secret and a public key stored on the server).



Why we use SSH?



1. Remote access:

You can open the server's terminal and type commands as if you were sitting at its keyboard.

2. Safe file transfers:

You can copy files between your computer and the server without worrying about hackers stealing them.

3. Port forwarding:

SSH can secretly carry other types of traffic inside its tunnel, protecting them too.

4. Data safety:

SSH makes sure your information isn't read, changed, or faked while traveling.



JS-Linux

JSLinux

Run Linux or other Operating Systems in your browser!

The following emulated systems are available:

CPU	OS	User Interface	VFsync access	Startup Link	TEMU Config	Comment
x86	Alpine Linux 3.12.0	Console	Yes	click here	url	
x86	Alpine Linux 3.12.0	X Window	Yes	click here	url	Right mouse button for the menu.
x86	Windows 2000	Graphical	No	click here	url	Disclaimer .
x86	FreeDOS	VGA Text	No	click here	url	
riscv64	Buildroot (Linux)	Console	Yes	click here	url	
riscv64	Buildroot (Linux)	X Window	Yes	click here	url	Right mouse button for the menu.
riscv64	Fedora 33 (Linux)	Console	Yes	click here	url	Warning: longer boot time.
riscv64	Fedora 33 (Linux)	X Window	Yes	click here	url	Warning: longer boot time. Right mouse button for the menu.

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Open link here: [JSLinux](#)

We'll use a browser-based Linux terminal to practice commands. This one runs on Alpine Linux, which is a lightweight version of Linux. The interface works just like any Linux terminal, so all the commands we need will behave the same way.



Overthewire level 0

```
localhost:~# ssh bandit0@bandit.labs.overthewire.org -p 2220
The authenticity of host '[bandit.labs.overthewire.org]:2220 ([56.228.72.241]:2220)' can't be established.
ECDSA key fingerprint is SHA256:IJ7FrX0mKSSHTJ63ezxjqtn0E0Hg116Aq+v5mN0+HdE.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '[bandit.labs.overthewire.org]:2220,[56.228.72.241]:2220' (ECDSA) to the list of known hosts.
```

The logo for OverTheWire, featuring the words "OverTheWire" in a stylized, blocky font where the letters are interconnected.

This is an OverTheWire game server.
More information on <http://www.overthewire.org/wargames>

```
backend: gibson-0
bandit0@bandit.labs.overthewire.org's password:
```

Bandit Level 0

Bandit is like a computer puzzle game.

To play, you connect to their computer using SSH. You type a magic key (username, password, server, and port) to enter.

Example:

```
ssh bandit0@bandit.labs.overthewire.org -p 2220
```

1. ssh → Open the secret tunnel.
2. bandit0@ → The name of the player (username).
3. bandit.labs.overthewire.org → The game's computer (server).
4. -p 2220 → The special door (port) we must use.

Note the password is: bandito



Using ls

```
bandit0@bandit:~$ ls  
readme
```

“ls” lists the files and folders in your current location.

Here we see document
readme —
nothing hidden, no details.

This is the fastest way to see
what's around you.



Using ls -a

-a = "all" → shows Visible items (normal files/folders you always see) and hidden items (names starting with a dot .).

```
bandit0@bandit:~$ ls -a
.  ..  .bash_logout  .bashrc  .profile  readme
bandit0@bandit:~$
```

A file or folder is considered hidden if its name begins with a dot (.).

These aren't "secret", they're just not shown by default because they're often system or application files you don't need to see all the time.



cat – show content

```
bandit0@bandit:~$ cat readme
Congratulations on your first steps into the bandit game!!
Please make sure you have read the rules at https://overthewire.org/rules/
If you are following a course, workshop, walkthrough or other educational activity,
please inform the instructor about the rules as well and encourage them to
contribute to the OverTheWire community so we can keep these games free!

The password you are looking for is: ZjLjTmM6FvvyRnrb2rfNW0Z0Ta6ip5If
```

cat stands for concatenate.

cat filename → shows the contents of a file.

You can use it to quickly peek at text files, notes, or passwords in Bandit.

cat readme — opens the file and shows its text. When we ran `cat readme`, the file revealed a congratulatory message and the password!

The Password is :

ZjLjTmM6FvvyRnrb2rfNW0Z0Ta6ip5If

Use that password to log into the next level (bandit1) using SSH on port 2220.



Bandit Level 1 to level 2

The password for next level is hiding in a file called - .

It lives in the home folder. Try to find it.

To play, you connect to their computer using SSH.
You type a magic key (username, password, server, and port) to enter.

Example:

```
ssh bandit1@bandit.labs.overthewire.org -p 2220
```

1. ssh → Open the secret tunnel.
2. bandit1@ → The name of the player (username) in this level.
3. bandit.labs.overthewire.org → The game's computer (server).
4. -p 2220 → The special door (port) we must use.

```
localhost:~# ssh bandit1@bandit.labs.overthewire.org -p 2220
```

```
bandit
```

```
This is an OverTheWire game server.  
More information on http://www.overthewire.org/wargames
```

```
backend: gibbon-0
```

```
bandit1@bandit.labs.overthewire.org's password:
```



ls commands

First task: find a file called `-` in the home folder.

That file holds the password for the next level.

Type:

1. `ls`: shows the files in the folder.
2. `ls -a`: shows everything, including hidden files and details.

```
bandit1@bandit:~$ ls
-
bandit1@bandit:~$ ls -a
- . . . .bash_logout .bashrc .profile
bandit1@bandit:~$
```



cat command

```
bandit1@bandit:~$ cat ./-  
263JGJPfgU6LtdEvgfWU1XP5yac29mFx
```

cat stands for concatenate.

cat filename → shows the contents of a file.

In this Bandit level, the password is hidden in a file called -.

That's a problem, because in Linux, - usually means "standard input" or an option flag.

There are a few tricks to tell Linux, "No, I really mean the file named -!"

Use the full path

cat ./-

Here, ./ means "the current folder," so Linux understands it's a file.



Bandit Level 2 to level 3

The password for the next level is stored in a file called:

- - spaces in this filename - -
located in the home directory.

To play, you connect to their computer using SSH.
You type a magic key (username, password, server, and port) to enter.

Example:

```
ssh bandit2@bandit.labs.overthewire.org -p 2220
```

1. ssh → Open the secret tunnel.
2. bandit2@ → The name of the player (username) in this level.
3. bandit.labs.overthewire.org → The game's computer (server).
4. -p 2220 → The special door (port) we must use.

```
localhost:~# ssh bandit2@bandit.labs.overthewire.org -p 2220
```

bandit

This is an OverTheWire game server.
More information on <http://www.overthewire.org/wargames>

```
backend: gibbon-0
```

```
bandit2@bandit.labs.overthewire.org's password:
```



ls commands

First task: find a file called
- - spaces in this filename - - in
the home folder.

That file holds the password for
the next level.

Type:

```
bandit2@bandit:~$ ls
--spaces in this filename--
bandit2@bandit:~$ ls -a
.  ..  .bash_logout  .bashrc  .profile  --spaces in this filename--
```

1. ls: shows the files in the folder.
2. ls -a: shows everything, including hidden files and details.



cat command

```
bandit2@bandit:~$ cat -- '--spaces in this filename--'  
MNk8KNH3Usiio41PRUEoDFPqfxLPISmx
```

In this level, the password is inside a file called: spaces in this filename

The problem is Linux normally treats spaces as separators between different words or commands.

So, we can handle spaces in file names in different ways:

Use quotes `cat - - '--spaces in this filename - -'`

`- -` tells Linux commands: "No more options after this, only filenames."



Bandit Level 3 to level 4

The password for the next level is stored in a hidden file in the inhere directory.

To play, you connect to their computer using SSH.

You type a magic key (username, password, server, and port) to enter.

Example:

```
ssh bandit3@bandit.labs.overthewire.org -p 2220
```

1. ssh → Open the secret tunnel.
2. bandit3@ → The name of the player (username) in this level.
3. bandit.labs.overthewire.org → The game's computer (server).
4. -p 2220 → The special door (port) we must use.

```
localhost:~# ssh bandit3@bandit.labs.overthewire.org -p 2220
```

bandit

This is an OverTheWire game server.
More information on <http://www.overthewire.org/wargames>

```
backend: gibbon-0
```

```
bandit3@bandit.labs.overthewire.org's password:
```



ls commands

First task: find in the inhere directory in the home folder.

That file holds the password for the next level.

1. ls: shows the files in the folder.
2. ls -a: shows everything, including hidden files and details.
3. When we add -l after ls, it switches to a "long listing" mode. Now, instead of just names, you get a full report about each item.

```
bandit3@bandit:~$ ls
inhere
bandit3@bandit:~$ ls -a
.  ..  .bash_logout  .bashrc  inhere  .profile
bandit3@bandit:~$ ls -l
total 4
drwxr-xr-x 2 root root 4096 Aug 15 13:16 inhere
```

So, we know that inhere is a directory as the first character is d so it's directory



cd command

Use cd to move between folders: In this example, we entered the inhere folder with: cd inhere

At first, ls and ls -l showed nothing.

But hidden files don't show up normally.

So, we tried: ls -a

And finally a hidden file appeared: ...Hiding-From-You

These commands help you explore the file system, which you'll often need to do when searching for logs, tools, or evidence during an investigation.

```
bandit3@bandit:~$ cd inhere
bandit3@bandit:~/inhere$ ls
bandit3@bandit:~/inhere$ ls -l
total 0
bandit3@bandit:~/inhere$ ls -a
.  ..  ...Hiding-From-You
bandit3@bandit:~/inhere$
```



cat command

In this level, the password is inside a file called: ...Hiding-From-You so, we use the command as follows:

cat ...Hiding-From-You → opened the file and displayed the password:

```
bandit3@bandit:~/inhere$ cat ...Hiding-From-You  
2WmrDFRmJIq3IPxneAaMGhap0pFhF3NJ
```

Password for the next level:
2WmrDFRmJIq3IPxneAaMGhapop
FhF3NJ



Bandit Level 4 to level 5

The password for the next level is stored in the only human-readable file in the inhere directory.
Tip: if your terminal is messed up, try the "reset" command.

To play, you connect to their computer using SSH. You type a magic key (username, password, server, and port) to enter.

Example:

```
ssh bandit4@bandit.labs.overthewire.org -p 2220
```

1. ssh → Open the secret tunnel.
2. bandit4@ → The name of the player (username) in this level.
3. bandit.labs.overthewire.org → The game's computer (server).
4. -p 2220 → The special door (port) we must use.

```
localhost:~# ssh bandit4@bandit.labs.overthewire.org -p 2220
```

The logo is a stylized representation of the word "OverTheWire" using a grid of characters. The characters are arranged in a way that they form the letters of the word, with some characters being white and others being black, creating a high-contrast, pixelated effect.

This is an OverTheWire game server.
More information on <http://www.overthewire.org/wargames>

```
backend: gibson-0
```

```
bandit4@bandit.labs.overthewire.org's password:
```



List files in directory

First task: find in the inhere directory in the home folder.

That file holds the password for the next level.

Type:

1. ls: shows the files in the folder.

2. Use cd to move between folders:
In this example, we entered the inhere folder with: cd inhere

```
bandit4@bandit:~$ ls
inhere
bandit4@bandit:~$ cd inhere
bandit4@bandit:~/inhere$ ls
-file00  -file02  -file04  -file06  -file08
-file01  -file03  -file05  -file07  -file09
```



cat command

We discovered that the files in this level were named with dashes:

-file00, -file01, -file02 ...

Such files are tricky because Bash treats names starting with - as options, not filenames.

So, we saw multiple files, each starting with a dash (-).

By using `cat -- *`, we read all the files at once.

Most were gibberish, but one line stood out clearly is the

Password:

4oQYVPkxZOOEOO5pTW81FB8j8lxXGUQw

```
bandit4@bandit:~/inhere$ ls
-file00 -file02 -file04 -file06 -file08
-file01 -file03 -file05 -file07 -file09
bandit4@bandit:~/inhere$ cat -- *
w%fykφ°KSvm²/DGou~XI'μv00`φRg.Ř²v0>0h ]rp0iL_ L´μJx".x0lxüm|0g莖TUû®uhjaW10dS
>0GycYQP%0P0ü;z`[;0üës00²o0xv,y±0010kG0Nÿ¥J4"¥R&1D0dM$100#1³nq)ň4oQYVPkxZOOEOO5p
TW81FB8j8lxXGUQw
2;ü0d$7 ,nF 90'u.¼0iI00¥V;C r00A<Dt>00μ`0bandit4@bandit:~/inhere$
```



Bandit Level 5 to level 6

The password for the next level is stored in a file somewhere under the inhere directory and has all of the following properties:

1. human-readable
2. 1033 bytes in size
3. not executable

To play, you connect to their computer using SSH. You type a magic key (username, password, server, and port) to enter.

Example:

`ssh bandit5@bandit.labs.overthewire.org -p 2220`

1. `ssh` → Open the secret tunnel.
2. `bandit5@` → The name of the player (username) in this level.
3. `bandit.labs.overthewire.org` → The game's computer (server).
4. `-p 2220` → The special door (port) we must use.

```
localhost:~# ssh bandit5@bandit.labs.overthewire.org -p 2220
```



This is an OverTheWire game server.
More information on <http://www.overthewire.org/wargames>

```
backend: gibson-0
```

```
bandit5@bandit.labs.overthewire.org's password:
```



List files in directory

First task: find in the inhere directory in the home folder.

That file holds the password for the next level.

Type:

1. ls: shows the files in the folder.
2. Use cd to move between folders: In this example, we entered the inhere folder with: cd inhere

```
bandit5@bandit:~$ ls
inhere
bandit5@bandit:~$ cd inhere
bandit5@bandit:~/inhere$ ls
maybehere00  maybehere04  maybehere08  maybehere12  maybehere16
maybehere01  maybehere05  maybehere09  maybehere13  maybehere17
maybehere02  maybehere06  maybehere10  maybehere14  maybehere18
maybehere03  maybehere07  maybehere11  maybehere15  maybehere19
bandit5@bandit:~/inhere$
```



find command

The file is human-readable (so file would say "ASCII text" or similar).

The file size is exactly 1033 bytes.

The file is not executable.

How to do that:

1. `maybehere*` — search all directories whose names start with `maybehere`.
2. `-type f` — only regular files (no directories).
3. `-size 1033c` — size exactly 1033 bytes (the `c` means bytes).! `-executable` — exclude executable files (keep only non-executable).
4. `-print` — print the path that matched.
5. `-exec file {} \;` — run file on the found file(s) so you can see if it's text (human-readable).
6. `-exec cat {} \;` — display the file contents (the password) immediately.

```
bandit5@bandit:~/inhere$ find maybehere* -type f -size 1033c ! -executable -print  
t -exec file {} \; -exec cat {} \;  
maybehere07/.file2  
maybehere07/.file2: ASCII text, with very long lines (1000)  
HWasnPhtq9AVKe0dmk45nxy20cvUa6EG
```



du command

find . → Start searching in the current directory.

-type f → Only look for files (ignore directories).

-size 1033c → Match files that are exactly 1033 bytes (c = bytes).!

-executable → Exclude executable files.

-exec du -b {} \; → For each file found, run du -b to display its size in bytes.

1. {} = placeholder for each found file

2. \; = ends the -exec command

```
bandit5@bandit:~/inhere$ find . -type f -size 1033c ! -executable -exec du -b {} \;  
1033    ./maybehere07/.file2
```

